Exchanging a Central Venous Catheter for a Vascular Catheter

This guide outlines the technique for exchanging an existing central venous catheter (CVC) for a vascular catheter (Vascath) for haemodialysis access.

In most cases, the existing CVC has a smaller lumen than the Vascath and therefore cannot accommodate the larger Vascath guidewire directly. Therefore, the aim of this technique is to use a peripheral intravenous (IV) cannula as a temporary conduit to facilitate insertion of the Vascath guidewire, allowing subsequent vessel dilation and Vascath insertion.

The key point is to select a peripheral IV cannula with an internal lumen large enough to accommodate the Vascath guidewire. It is **critical** to check this at the start of the procedure. The cannula sizes suggested below serve as a guide, but the internal diameter may change according to different manufacturers, so please check before the start of the procedure.

Procedure

Perform the procedure using sterile technique.

1. Insert an appropriately sized CVC guidewire through the distal port of the existing CVC.
    - A replacement 0.018” CVC guidewire is available separately and can be used without opening a new CVC set for 4Fr - 5.5Fr CVCs.
2. While maintaining guidewire position within the vein, remove the existing CVC.
3. After removing the CVC, maintain pressure to minimize bleeding.
4. Advance the peripheral IV cannula over the CVC guidewire into the vessel.
5. Leave the cannula in situ and carefully remove the CVC guidewire.
    - On occasion, serially upsizing with a larger IV cannula and larger guidewire may be needed to accommodate the Vascath guidewire.
6. Insert the Vascath guidewire through the peripheral IV cannula.
7. Remove the peripheral IV cannula, leaving the Vascath guidewire in place.
8. Dilate the vessel using the dilators provided in the Vascath kit.
9. Insert the Vascath according to the standard insertion technique.

Essential:

- Ensure that you maintain control of the guidewire at all times throughout the procedure.
- Never force or kink a guidewire, it should be able to pass smoothly and without resistance.

Commonly used CVC/Vascath and Guidewire sizes:

CVC size (French)	Wire Size (diameter)
4	0.018” (0.46mm)
4.5	0.018” (0.46mm)
5.5	0.018” (0.46mm)
7	0.032” (0.81mm)

Vascath size (French)	Wire Size (diameter)	Peripheral Cannula size (B Braun Vasofix safety)
6.5	0.028” (0.71mm)	20G (ID 0.76mm)
8	0.035” (0.89mm)	18G (ID 0.96mm)
11.5	0.032” (0.81mm)	18G (ID 0.96mm)
13	0.032” (0.81mm)	18G (ID 0.96mm)

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